Introduction to Mobile IPv6

IIS5711: Mobile Computing

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Outline

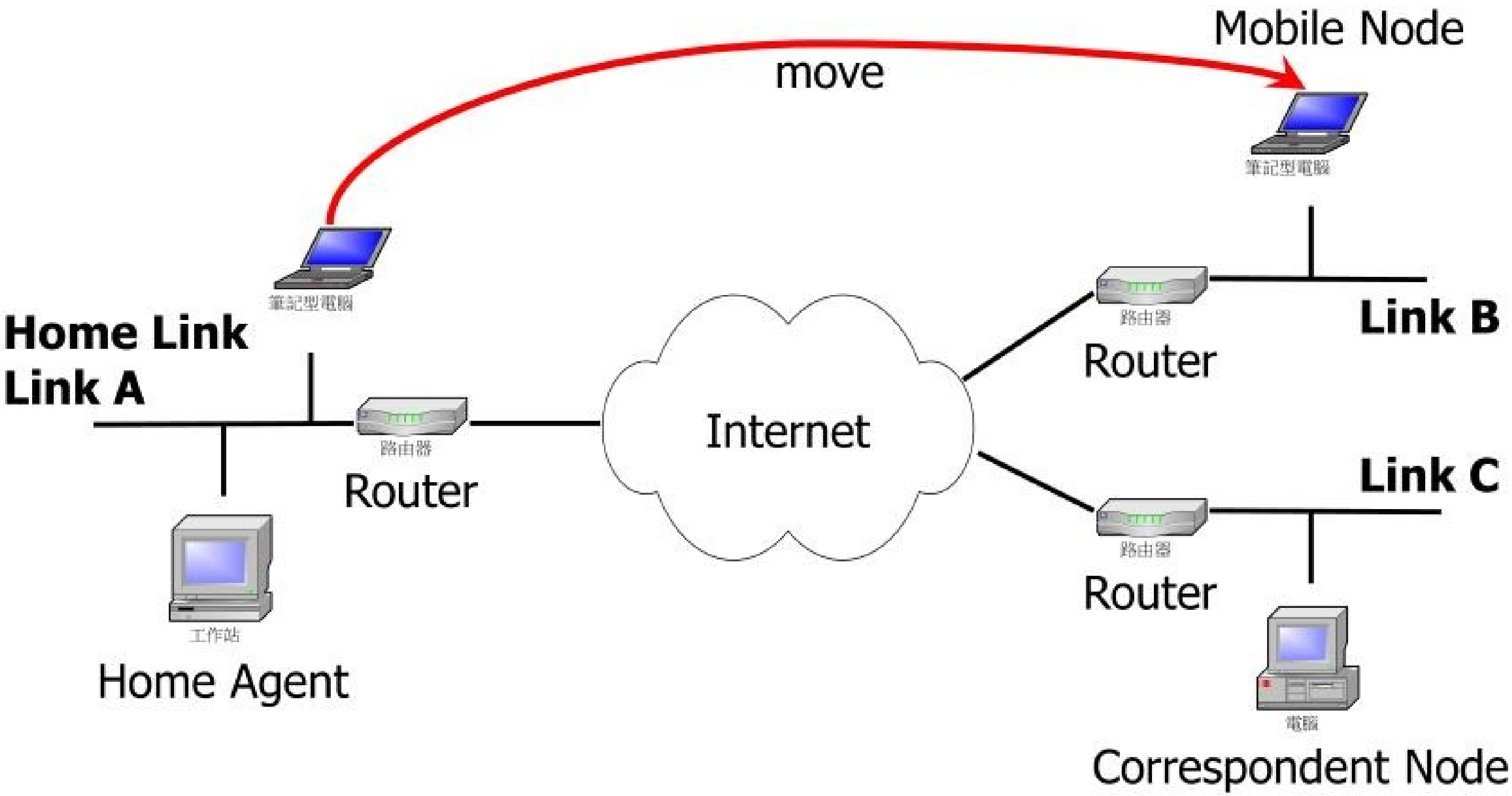
- Introduction
- Relevant Features of IPv6
- Major Differences between MIPv4 and MIPv6
- Mobile IPv6 Operation
- Home Agent Discovery Mechanism
- Handover
- Quality of Service
- Conclusions
- References

Introduction

- Mobile IPv6 is intended to enable IPv6 nodes to move from one IP subnet to another
- While a mobile node is away from home
  - It sends information about its current location to a home agent
  - The home agent intercepts packets addressed to the mobile node and tunnels them to the mobile node's present location

Introduction (cont.)

- Mobile IPv6 scenario



Relevant Features of IPv6

- Address Autoconfiguration
  - Stateless autoconfiguratoin
    - Network Prefix + Interface ID
  - Stateful autoconfiguration
    - DHCPv6
- Neighbor Discovery
  - Discover each other's presence and find routers
  - Determine each other's link-layer addresses
  - Maintain reachability information

Relevant Features of IPv6 (cont.)

- Extension Headers
  - Routing header
    - For route optimization
  - Destination Options header
    - For mobile node originated datagrams

Major Differences between MIPv4 and MIPv6

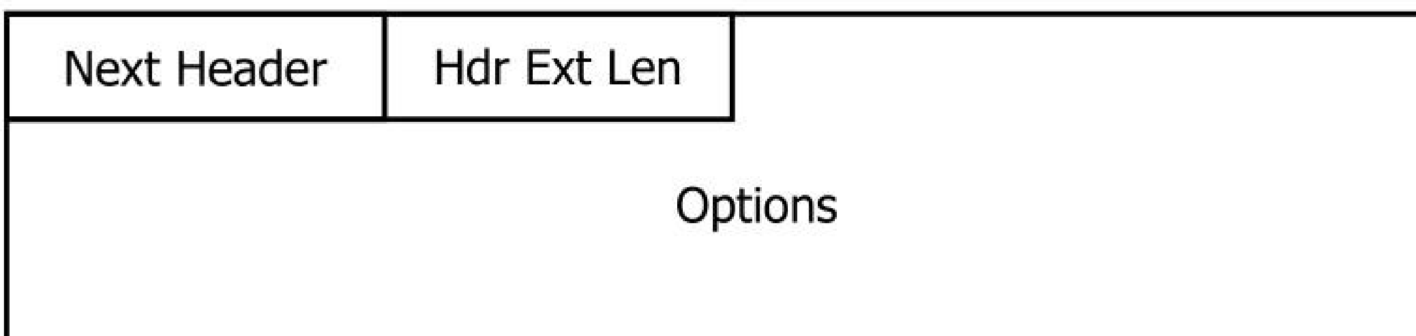
- No FA in Mobile IPv6
  - Mobile IPv6 requires every mobile node to support
    - IPv6 Decapsulation
    - Address Autoconfiguration
    - Neighbor Discovery

Major Differences between MIPv4 and MIPv6 (cont.)

- Packets delivery
  - MIPv6 mobile node uses care-of address as source address in foreign links
    - No ingress filtering problem
  - Correspondence Node uses IPv6 routing header rather than IP encapsulation
    - Supports "Route Optimization" naturally

Mobile IPv6 Messages and Related Data Structures

- All new messages used in MIPv6 are defined as IPv6 Destination Options
  - These options are used in IPv6 to carry additional information that needs to be examined only by a packet's destination node



Mobile IPv6 Messages and Related Data Structures (cont.)

- Four new Destination Options
  - Binding Update
    - Used by an MN to inform its HA or any other CN about its current care-of address
  - Binding Acknowledgement
    - Used to acknowledge the receipt of a Binding Update

Mobile IPv6 Messages and Related Data Structures (cont.)

- Binding Request
  - Used by any node to request an MN to send a Binding Update with the current care-of address
- Home Address
  - Used in a packet sent by a mobile node to inform the receiver of this packet about the mobile node's home address

Mobile IPv6 Messages and Related Data Structures (cont.)

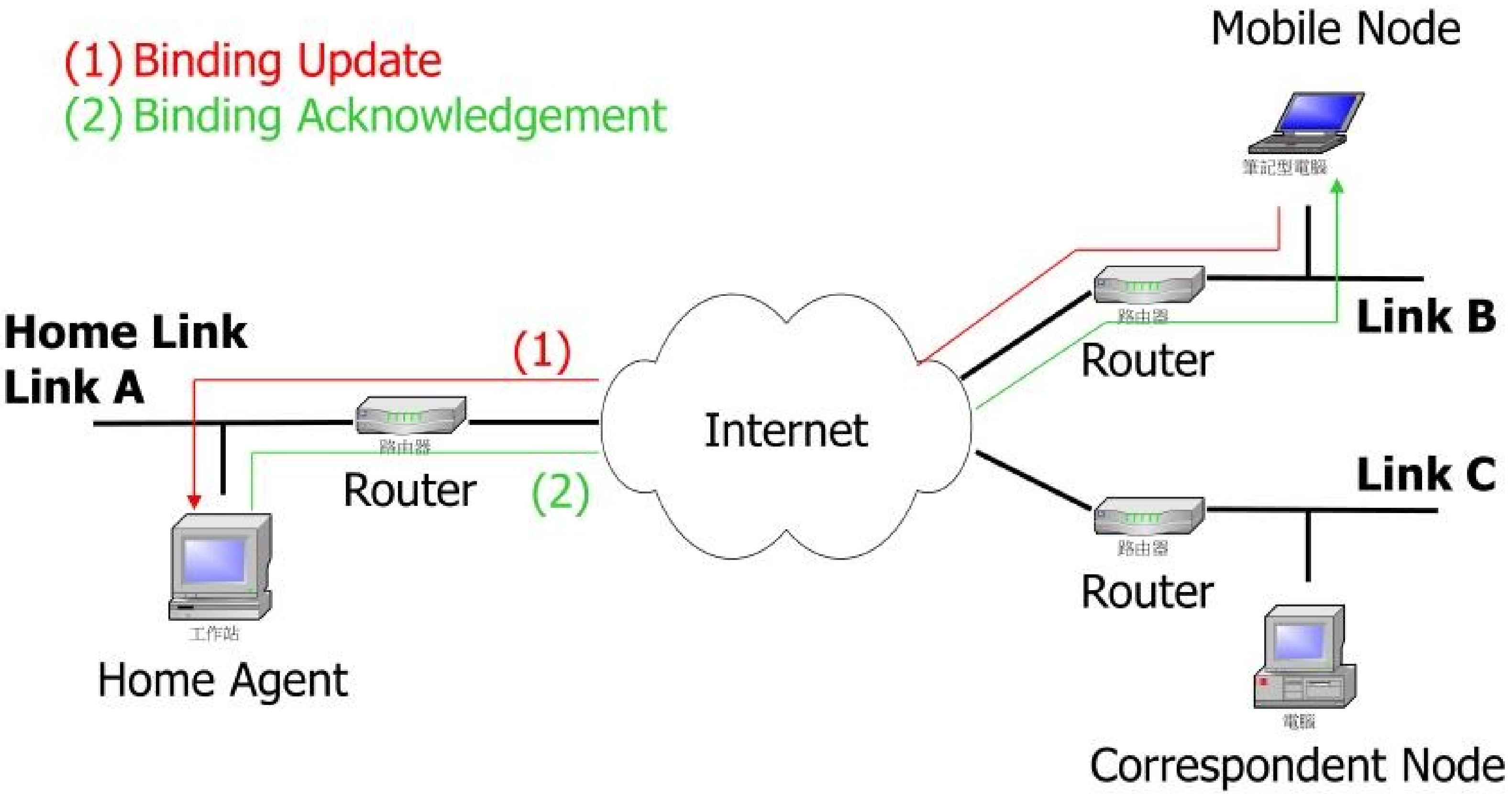
- Data Structures
  - Binding Cache
  - Binding Update List
  - Home Agent List

Mobile IPv6 Operation

- Home Agent Registration
  - An MN performs address autoconfiguration (stateful or stateless) to get its care-of address
  - The MN registers its care-of address with its home agent on the home link
    - Use “Binding Update” Destination Option
  - The HA uses proxy Neighbor Discovery and also replies to Neighbor Solicitations on behalf of the MN

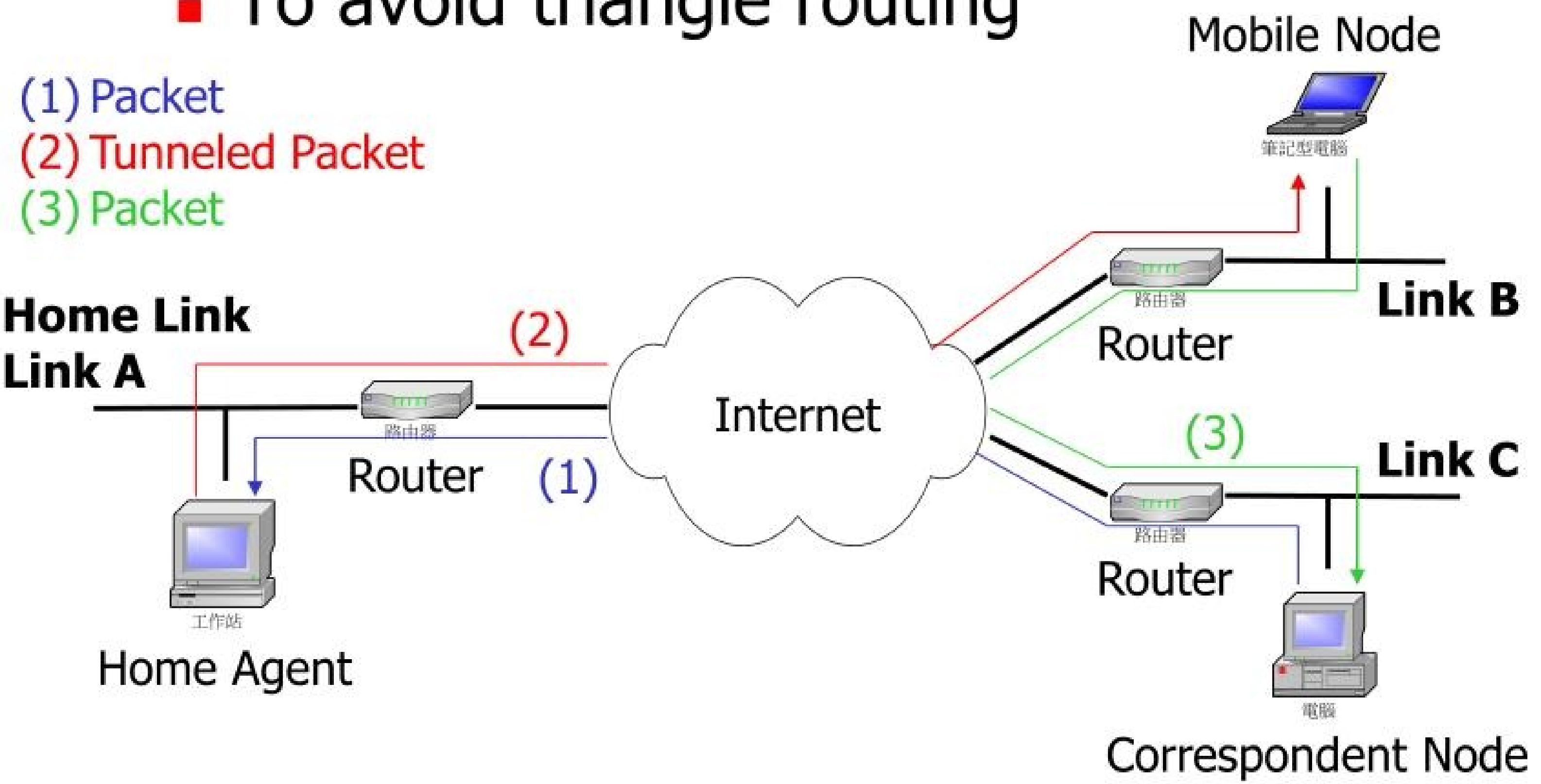
Mobile IPv6 Operation (cont.)

- Home Agent Registration



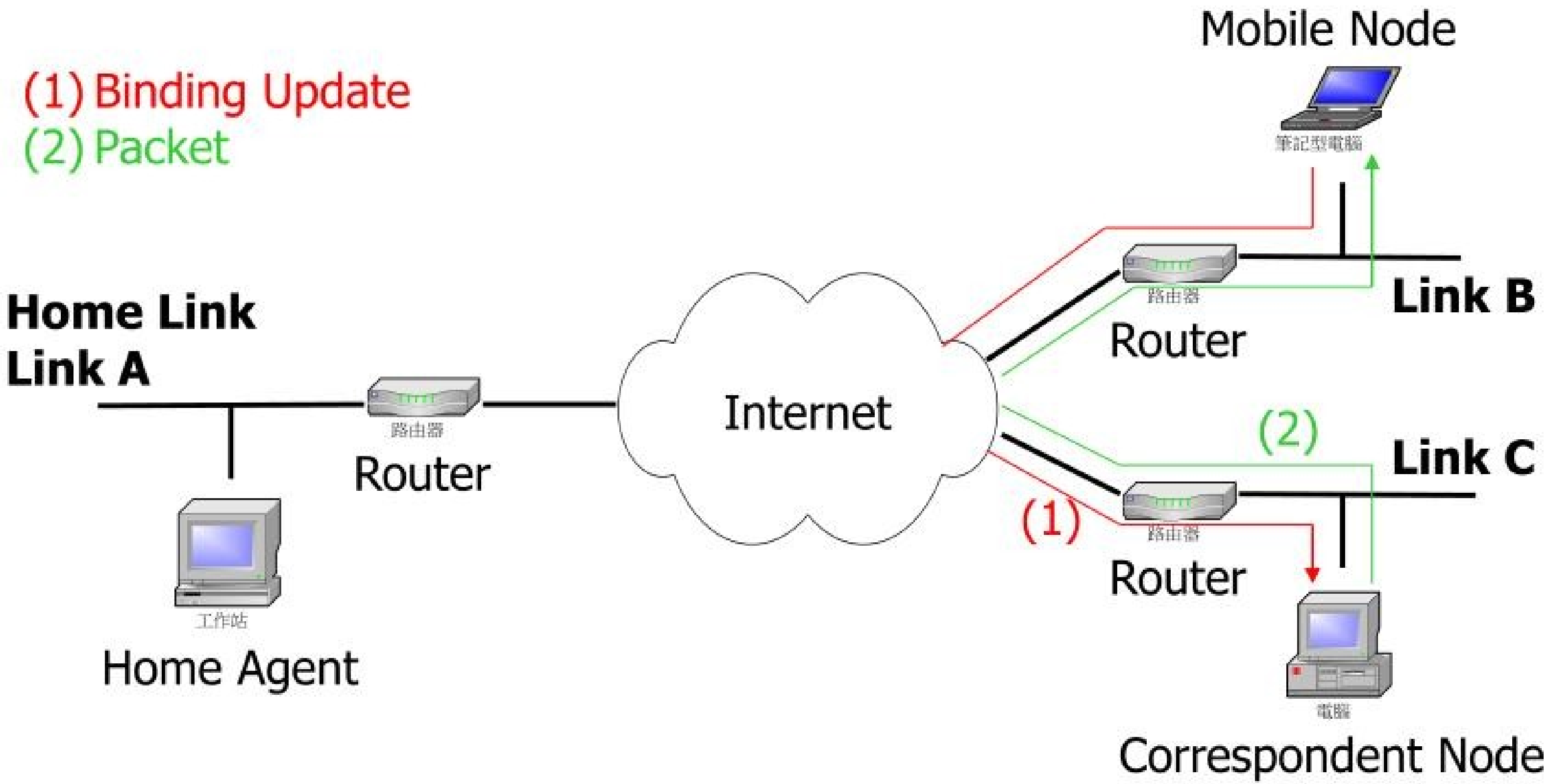
Mobile IPv6 Operation (cont.)

- Route Optimization
  - To avoid triangle routing



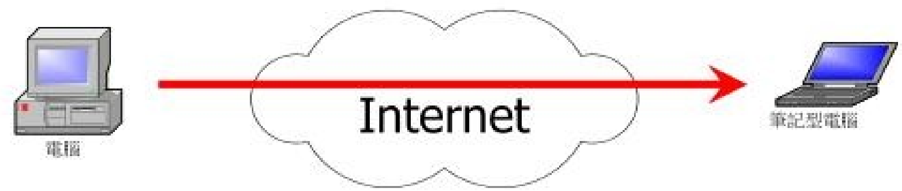
Mobile IPv6 Operation (cont.)

- Route Optimization



Mobile IPv6 Operation (cont.)

- MN-Terminated Packet Delivery



Src: CN's address
Dst: MN's home address

↓ Binding cache hit

Src: CN's address
Dst: MN's care-of address
Routing Header:
MN's home address

→

Src: CN's address
Dst: MN's care-of address
Routing Header:
MN's home address

↓

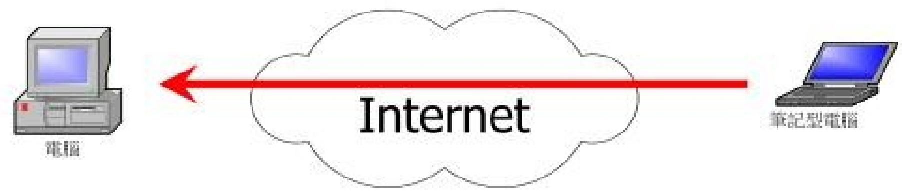
Src: CN's address
Dst: MN's home address

↓

MN receives packets from loopback interface

Mobile IPv6 Operation (cont.)

- MN-Originated Packet Delivery



MN at home:

Src: MN's home address
Dst: CN's address

MN at visited network:

Src: MN's care-of address
Dst: CN's address

Destination Optoins header –
Home Address Option:
MN's home address

Src: MN's care-of address
Dst: CN's address

Destination Optoins header –
Home Address Option:
MN's home address

Move MN's home address to Source Address

Src: MN's home address
Dst: CN's address

Mobile IPv6 Operation (cont.)

- Movement Detection
  - While away from home, an MN selects one router and one subnet prefix advertised by that router to use as the subnet prefix in its primary care-of address
  - To wait for the periodically sent Router Advertisements

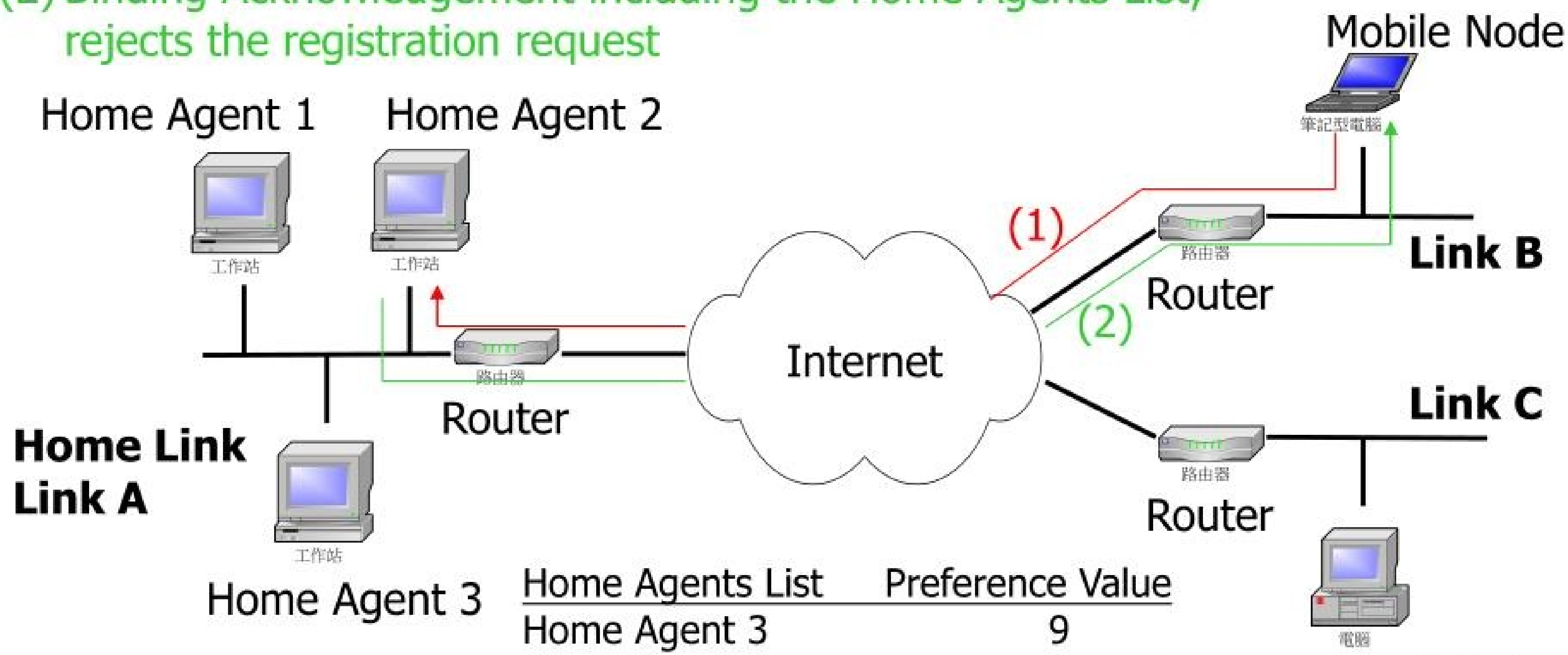
Mobile IPv6 Operation (cont.)

- Binding Management
  - To trigger Binding Acknowledgement, the MN sets the Acknowledge bit in the Binding Update
    - Retransmitting the Biding Update periodically until receipt of the acknowledgement
  - An MN MUST set the Acknowledge bit in Binding Updates addressed to an HA
  - The MN MAY also set the Acknowledge bit in Binding Updates sent to a CN

Home Agent Discovery Mechanism

(1) Binding Update to Home-Agents anycast address

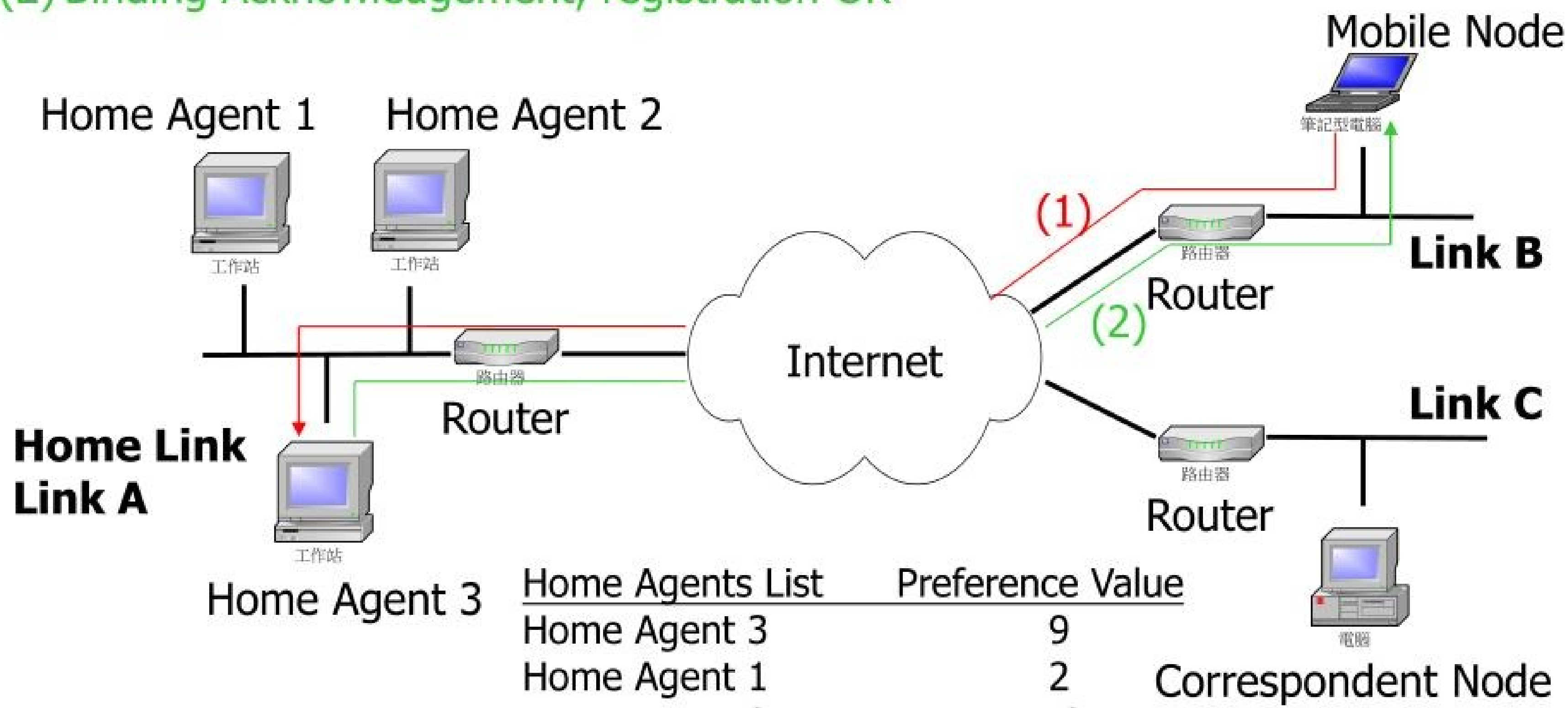
(2) Binding Acknowledgement including the Home Agents List; rejects the registration request



Home Agents List	Preference Value
Home Agent 3	9
Home Agent 1	2
Home Agent 2	-3

Home Agent Discovery Mechanism (cont.)

(1) Binding Update to Home Agents 3
(2) Binding Acknowledgement, registration OK



Home Agents List	Preference Value
Home Agent 3	9
Home Agent 1	2
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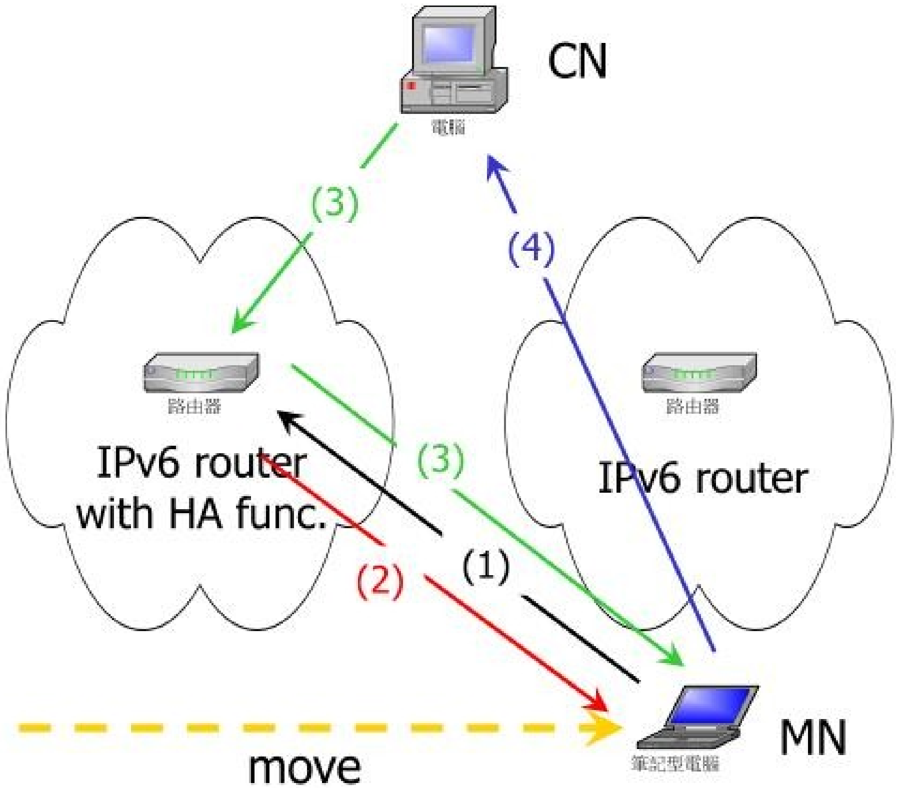
Handover

- Router-Assisted Smooth Handovers



(1) MN sends a Binding Update to an HA on previous network

(2) HA returns a Binding Acknowledgement

(3) HA tunnels packets to MN

(4) MN sends a Binding Update to CN

Handover (cont.)

- Three kinds of handover operations
  - Smooth Handover
    - Minimizes data loss during the time that the MN is establishing its link to the new access point
  - Fast Handover
    - Minimizes or eliminates latency for establishing new communication paths to the MN at the new access router
  - Seamless Handover
    - Both Smooth and Fast Handover

Quality of Service

- IPv6 header has two QoS-related fields
  - 20-bit Flow Label
    - Used by a source to label sequences of packets for which it requests special handling by the IPv6 routers
    - Geared to IntServ and RSVP
  - 8-bit Traffic Class Indicator
    - Used by originating nodes and/or forwarding routers to identify and distinguish between different classes or priorities of IPv6 packets
    - Geared to DiffServ

Quality of Service (cont.)

- New IPv6 option – QoS Object
  - QoS Object describes QoS requirement, traffic volume and packet classification parameters for MN's packet stream
  - Included as a Destination Option in IPv6 packets carrying Binding Update and Biding Acknowledgment messages

Conclusions

- Mobile IPv6 is
  - An efficient and deployable protocol for handling mobility with IPv6
  - Lightweight protocol
  - To minimize the control traffic needed to effect mobility

References

- C. Perkins, "Mobility for IPv6," *Internet Draft*, June 2002.
- K. Zhigang et al., "QoS in Mobile IPv6," in *Proc. of International Conferences on Info-tech and Info-net 2001*, vol. 2, pp. 492 -497.
- N. Montavont and T. Noel, "Handover Management for Mobile Nodes in IPv6 Networks," *IEEE Communication Magazine*, pp. 38-43, Aug. 2002.